Healthcare Capacity, Care Load and Flow Dynamics in the U.S. Unaccompanied Alien Children Program: An Exploratory Data Analysis and Operational Analytics Study

Abstract

The Unaccompanied Alien Children (UAC) Program operates as a dynamic child-welfare and care-delivery pipeline in which children referred by federal immigration authorities enter the care network, receive medical, mental-health, educational, case-management, and other services, and are ultimately discharged to vetted sponsors or other appropriate placements. The Administration for Children and Families (ACF) notes that the Office of Refugee Resettlement (ORR) uses referrals, projections, and estimates to inform program resources, including bed-capacity needs, and emphasizes flexible capacity and community-based care.

This study analyzes daily UAC operational data covering January 12, 2023 through December 21, 2025. The dataset contains measures of children apprehended and placed in CBP custody, children in CBP custody, transfers from CBP custody, children in HHS care, and HHS discharges.

After removal of completely blank records, 720 observations remained from the 1,170-row CSV. The analysis identifies substantial changes in system load across the study period. Average total system load was approximately 6,233 children, while the maximum observed combined CBP and HHS load reached 11,762 children on December 20, 2023. Average HHS care load was approximately 6,061 children, considerably larger than the average CBP custody population of approximately 171 children.

Flow analysis shows that transfers and discharges were not consistently balanced. Across observed reporting days, 238 days recorded positive net intake pressure, while 475 recorded negative net pressure. The overall average daily net pressure was approximately -44.7 children, indicating that, across the observed reporting days, discharges generally exceeded transfers. However, this overall result masks periods of substantial accumulation and relief.

The study recommends a continuous operational monitoring framework based on total system load, net intake pressure, rolling pressure indicators, backlog accumulation, reporting completeness, and anomaly monitoring. Because the dataset does not contain explicit facility-capacity denominators, the analysis distinguishes **care-load pressure** from actual **facility overcapacity**. The resulting framework can support staffing, shelter planning, resource allocation, and proactive humanitarian response without overstating what the available data can establish.

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1. Introduction

The UAC Program represents a complex federal care-delivery system operating at the intersection of child welfare, healthcare, immigration-related referral processes, shelter operations, family reunification, and sponsor placement.

The operational pipeline can be represented as:

CBP apprehension → CBP custody → transfer to HHS/ORR care → medical and welfare services → discharge to sponsor or other placement.

ORR does not apprehend children at the border; rather, children are referred to ORR by the Department of Homeland Security. ORR uses referral information and projections to plan resources, including bed-capacity requirements.

This makes operational analytics particularly important. A sudden increase in referrals may not immediately appear as an equivalent increase in HHS care. Similarly, a high number of discharges can reduce the active care population even while incoming transfers remain substantial.

Consequently, a single metric such as daily apprehensions is insufficient to understand system pressure.

This study therefore treats the UAC Program as a **dynamic care pipeline** and evaluates:

- active care load;
- movement into HHS care;
- movement out of HHS care;
- changes in system-wide responsibility;
- sustained periods of positive or negative flow pressure;
- reporting completeness; and
- operational data-quality anomalies.

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2. Problem Statement

Operational data alone does not automatically provide operational intelligence.

Without a structured analytical framework, decision-makers may have difficulty determining:

1. whether the care system is accumulating children;
2. whether current pressure is temporary or sustained;
3. whether the CBP or HHS component carries most of the active workload;
4. whether discharge activity is sufficiently offsetting transfers;
5. whether apparent changes represent real operational changes or reporting gaps; and
6. when additional planning attention should be triggered.

This creates a risk of reactive rather than proactive capacity planning.

The central research problem is therefore:

> **How can daily UAC operational data be transformed into a healthcare-capacity monitoring framework capable of identifying changes in care load, intake pressure, backlog accumulation, and operational stress?**

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3. Research Objectives

3.1 Primary Objectives

The study seeks to:

- quantify daily and cumulative care load across CBP and HHS;
- measure the balance between transfers into HHS care and HHS discharges;
- identify periods of elevated system pressure;
- analyze temporal changes in the care population; and
- establish a repeatable KPI framework for operational monitoring.

3.2 Secondary Objectives

The study also aims to:

- support staffing and shelter planning;
- improve situational awareness for policymakers;
- identify reporting-quality limitations;
- distinguish temporary fluctuations from sustained pressure;
- provide a foundation for forecasting; and
- create a dashboard-ready analytical framework.

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4. Dataset Description

The uploaded CSV contains six principal operational variables:

Variable	Description
Date	Reporting date
Children apprehended and placed in CBP custody	Daily intake/apprehension measure
Children in CBP custody	Active CBP custody population

Variable	Description
Children transferred out of CBP custody	Flow from CBP into the HHS system
Children in HHS Care	Active HHS care population
Children discharged from HHS Care	Children leaving HHS care

These fields establish a useful flow representation from initial apprehension through custody, transfer, HHS care, and discharge.

The raw CSV contains 1,170 rows. After removing completely blank rows, 720 populated observations were available for analysis. The reporting period extends from January 12, 2023 to December 21, 2025.

A major analytical consideration is that the observations do not constitute a complete daily calendar. There are 355 calendar dates without a populated observation between the first and last reporting dates. These gaps are treated as **unreported/missing observations rather than zero operational activity**.

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5. Methodology

5.1 Data Cleaning

The analysis followed these steps:

1. Load the CSV.
2. Remove completely blank records.
3. Convert Date to a datetime object.
4. Remove comma separators from numeric HHS values.
5. Convert operational fields to numeric types.
6. Sort observations chronologically.
7. test for duplicate dates.
8. Construct a complete daily calendar.
9. identify missing reporting dates.
10. calculate derived operational metrics.
11. flag logical inconsistencies without automatically deleting them.

This approach preserves operational observations while maintaining transparency around data limitations.

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6. Analytical Framework

6.1 Total System Load

The principal workload indicator is:

$$Total\ System\ Load = CBP\ Custody + HHS\ Care$$

This measures the number of children represented in the two active care stages captured by the dataset.

6.2 Net Intake Pressure

The HHS-side flow-pressure measure is:

$$Net\ Intake\ Pressure = Transfers\ to\ HHS - HHS\ Discharges$$

Interpretation:

- **Positive value:** more children transferred into HHS care than discharged.
- **Negative value:** discharges exceed transfers.
- **Zero:** transfers and discharges are balanced.

A positive value should not automatically be interpreted as a crisis. It becomes operationally more meaningful when it persists for multiple reporting periods.

6.3 Care Load Growth

Daily change is:

$$Daily\ Load\ Change = Load_t - Load_{t-1}$$

The percentage growth rate is:

$$Growth_t = \frac{Load_t - Load_{t-1}}{Load_{t-1}} \times 100$$

These indicators identify sudden changes in system responsibility.

6.4 Rolling Pressure

Seven-day and fourteen-day rolling averages were used to reduce the effect of individual-day fluctuations.

$$Pressure_{7d} = Mean(Net\ Intake\ Pressure_{t-6:t})$$

$$Pressure_{14d} = Mean(Net\ Intake\ Pressure_{t-13:t})$$

A sustained positive rolling value provides stronger evidence of accumulation than a single positive observation.

6.5 Backlog Accumulation

A simple analytical backlog-pressure indicator was calculated as cumulative positive net intake:

$$Backlog = \sum max(Net\ Intake\ Pressure, 0)$$

This should be interpreted as a **pressure indicator**, not as a literal count of children awaiting placement, because the available dataset does not provide individual-level waiting-time or placement-status information.

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7. Exploratory Data Analysis

7.1 Overall System Load

Across the 720 populated observations:

- Mean total system load: approximately **6,233 children**.
- Median total system load: approximately **6,634 children**.
- Minimum observed total system load: approximately **2,002 children**.
- Maximum observed total system load: **11,762 children**.
- Maximum system load occurred on **December 20, 2023**.

At the peak observation, the combined population consisted of approximately 246 children in CBP custody and 11,516 children in HHS care.

The magnitude of the difference between the CBP and HHS populations demonstrates that the HHS component is the dominant component of the observed active-care workload.

7.2 CBP Versus HHS Care

Average CBP custody was approximately:

171 children per reporting day

while average HHS care was approximately:

6,061 children per reporting day.

Therefore, the observed care burden is overwhelmingly concentrated in HHS care rather than the CBP custody component.

This has an important operational implication: monitoring only border apprehensions or CBP custody would provide an incomplete picture of the care system's workload.

The HHS care population should therefore be treated as a primary operational indicator.

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8. Year-over-Year Analysis

The annual comparison demonstrates a substantial change in observed system load.

Year	Average Total Load	Maximum Load	Average HHS Care	Average CBP Custody
2023	8,847	11,762	8,646	200
2024	7,320	10,994	7,043	277
2025*	2,576	6,471	2,543	33

*2025 represents observations through December 21, rather than a complete calendar year.

The average observed system load declined substantially from 2023 to 2024 and again in the available 2025 observations.

This indicates that the UAC care environment was not characterized by a constant workload. Instead, the system experienced significant shifts in the size of the active care population.

For operational planning, this supports a **flexible-capacity model** rather than a static assumption of constant demand.

ORR's own planning framework emphasizes adaptable capacity and the use of referral projections to inform resource requirements.

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9. Flow and Intake Analysis

Average daily transfer activity was approximately:

128.7 children

while average daily HHS discharge activity was approximately:

173.4 children.

The overall average net intake pressure was therefore approximately:

-44.7 children per reporting day.

This suggests that, across the available observations, discharge activity generally exceeded transfers.

However, this aggregate statistic hides important variation.

Positive-pressure observations

- Positive net pressure: **238 days**
- Negative net pressure: **475 days**
- Zero net pressure: **7 days**

Thus, the system experienced positive intake pressure on roughly one-third of observed reporting days.

This is operationally important because even when the long-run average is negative, short periods of sustained positive pressure can temporarily increase HHS workload.

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10. Pressure Interpretation

The data demonstrate why a simple average is insufficient for operational decision-making.

A system may have:

- a negative annual average;
- several weeks of positive pressure;
- rising HHS care;
- increasing cumulative backlog pressure; and
- elevated volatility.

In such a situation, the overall annual average could falsely suggest that the system is comfortable.

Therefore, the recommended monitoring hierarchy is:

Daily pressure → 7-day pressure → 14-day pressure → sustained-pressure duration → system-load trend.

This allows policymakers to distinguish:

Short-term spike

One or two unusually high transfer days.

Emerging pressure

Several consecutive days of positive net intake.

Sustained pressure

Positive rolling averages combined with rising system load.

Relief period

Persistent negative net intake accompanied by declining system load.

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11. Data Quality Analysis

Data quality is a central finding of this study.

11.1 Blank Records

The raw CSV contains 1,170 rows, but 450 are completely blank.

These records should be removed before analysis.

11.2 Missing Calendar Dates

The populated dataset contains 720 reporting observations across a calendar span that contains substantially more than 720 days.

The resulting **355 missing calendar dates** should not be interpreted as days with zero children.

Instead, the dashboard should explicitly identify:

No reported observation

rather than:

Zero activity.

This distinction is critical for time-series analysis.

11.3 Transfer Consistency

The requested logical check was:

$$Transfers \leq CBP\ Custody$$

The dataset contains **86 observations where transfers exceed the same-day CBP custody count**.

These records should be flagged rather than automatically deleted.

A same-day comparison may not represent a strict stock-flow relationship because custody is a point-in-time stock while transfers represent activity during a reporting period.

Therefore, the appropriate classification is:

Temporal/operational consistency flag

rather than:

Confirmed data error.

11.4 Discharge Consistency

No observations were identified in which:

$$Discharges > HHS\ Care$$

This provides stronger internal consistency for the HHS stock-flow relationship.

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12. Care Load Volatility

The standard deviation of daily changes in total system load was approximately:

140 children.

This indicates meaningful day-to-day variation in the observed system population.

The operational implication is that staffing and shelter planning based solely on average load may underestimate short-term demand fluctuations.

A more resilient approach would combine:

- average load;
- peak load;
- rolling load;
- daily volatility; and
- recent flow pressure.

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13. Key Insights

Insight 1 — HHS care is the dominant workload component

The average HHS population is substantially larger than the average CBP custody population.

Implication: HHS care capacity should be a central focus of operational monitoring.

Insight 2 — System pressure is dynamic

The dataset includes both accumulation and relief periods.

Implication: Static annual averages are insufficient for real-time operational planning.

Insight 3 — 2023 represented the highest observed workload environment

The maximum observed total system load reached 11,762 children in December 2023.

Implication: Historical peak periods should be incorporated into contingency planning and scenario analysis.

Insight 4 — Discharges frequently offset transfers

Across the observed reporting days, negative net pressure was considerably more common than positive pressure.

Implication: Discharge throughput is a major mechanism for preventing prolonged accumulation.

Insight 5 — Short-term pressure can exist even when aggregate pressure is negative

The overall average net pressure is negative, but 238 reporting observations show positive pressure.

Implication: Operational dashboards should prioritize rolling and sustained-pressure indicators rather than relying solely on cumulative averages.

Insight 6 — Reporting completeness must be part of the dashboard

The presence of 355 missing calendar dates means that trends should always be interpreted alongside data-coverage indicators.

Implication: A data-quality panel should be treated as an operational feature rather than a technical afterthought.

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14. Healthcare and Operational Recommendations

Recommendation 1 — Implement a rolling capacity-pressure monitoring system

HHS operational leadership should monitor:

- current HHS census;
- total system load;
- 7-day net pressure;
- 14-day net pressure;
- recent peak load;
- discharge throughput; and
- sustained positive-pressure duration.

A rolling-pressure approach can identify emerging stress earlier than monthly averages.

Recommendation 2 — Use flexible capacity planning

The substantial variation between 2023, 2024, and 2025 demonstrates that a fixed staffing or shelter model may be inefficient.

ORR has emphasized the importance of adaptable standard shelter capacity and community-based care options when census levels change.

Planning should therefore include:

- baseline capacity;
- surge capacity;
- contingency capacity;
- staffing escalation thresholds; and
- demobilization thresholds.

Recommendation 3 — Monitor discharge throughput as closely as intake

Transfers receive considerable attention because they represent incoming workload.

However, this analysis shows that discharges are equally important to system stability.

A useful operational KPI is:

$$Discharge\ Offset\ Ratio = \frac{Discharges}{Transfers}$$

Interpretation:

- 1: discharge throughput exceeds transfer activity.

- =1: approximate flow balance.
- <1: incoming transfers exceed discharges.

This metric should be monitored alongside HHS census.

Recommendation 4 — Establish a formal pressure-alert framework

A possible analytical alert structure is:

Green

7-day pressure $\leq$ 0 and system load stable/declining.

Yellow

7-day pressure > 0 for several consecutive observations.

Orange

14-day pressure > 0 and total system load increasing.

Red

Sustained positive pressure combined with a historically high system load.

These thresholds should be calibrated using historical distributions and, most importantly, actual facility capacity data before being adopted as official operational thresholds.

Recommendation 5 — Integrate actual capacity data

The current dataset measures workload but not physical capacity.

Future data collection should include:

- available beds;
- occupied beds;
- staffing capacity;
- healthcare staffing ratios;
- facility type;
- geographic location;
- temporary surge capacity;
- average length of stay;
- placement wait time;
- sponsor approval time; and
- facility-level utilization.

This would allow the analysis to move from **capacity pressure monitoring** to genuine **capacity utilization forecasting**.

Recommendation 6 — Improve reporting completeness

The 355 missing calendar dates make continuous trend analysis difficult.

A centralized reporting pipeline should distinguish among:

1. zero activity;
2. missing report;
3. delayed report;
4. corrected report; and
5. operational anomaly.

This would substantially improve the reliability of forecasting and automated alerts.

Recommendation 7 — Develop a predictive early-warning model

Future work should forecast:

- HHS census;
- transfer volume;
- discharge volume;
- probability of sustained pressure; and
- probability of reaching historical high-load conditions.

A forecasting system should provide a range rather than a single point prediction.

For example:

Expected HHS care: 2,600

with an uncertainty interval such as:

2,350–2,900

rather than presenting the estimate as certain.

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15. Streamlit Decision-Support Dashboard

The research findings support development of a dashboard with five principal modules.

Module 1 — System Load Overview

Display:

- Total Children Under Care
- CBP Custody
- HHS Care
- Historical Peak
- Current trend

Module 2 — Intake and Discharge Flow

Display:

- Transfers
- Discharges
- Net Intake Pressure
- Discharge Offset Ratio

Module 3 — Pressure Monitoring

Display:

- daily pressure;
- 7-day rolling pressure;
- 14-day rolling pressure;
- sustained positive-pressure duration.

Module 4 — Historical Analysis

Allow users to switch between:

- daily;
- weekly; and
- monthly views.

Module 5 — Data Quality

Display:

- reporting coverage;
- missing dates;
- duplicate dates;
- transfer anomalies;
- discharge anomalies.

This design ensures that decision-makers can see both the operational situation and the reliability of the underlying data.

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16. Limitations

Several limitations should be recognized.

16.1 No facility-capacity denominator

The dataset does not specify the number of available beds or facility-level capacity.

Therefore, the analysis measures **care load** and **flow pressure**, not actual capacity utilization.

16.2 Missing reporting dates

The presence of 355 missing calendar dates reduces the reliability of continuous time-series modeling.

16.3 Aggregate data

The dataset does not contain child-level information.

Therefore, it cannot directly evaluate:

- individual length of stay;

- medical outcomes;
- demographic disparities;
- placement outcomes by child;
- case complexity; or
- individual-level healthcare needs.

16.4 Same-day stock-flow comparisons

A transfer is a flow, whereas CBP custody is a stock measured at a point in time.

Consequently, a transfer exceeding same-day custody should not automatically be treated as an erroneous record.

16.5 No causal inference

The analysis identifies patterns and associations.

It does not establish that a particular policy, border event, staffing change, or operational intervention caused changes in care load.

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17. Future Research

Future research should expand the dataset and analytical framework in several directions.

17.1 Facility-level analysis

Facility-level data would allow:

- utilization rates;
- geographic pressure maps;
- facility bottleneck detection;
- staffing optimization; and
- regional surge planning.

17.2 Length-of-stay analysis

Individual-level entry and discharge timestamps would allow survival analysis and identification of factors associated with extended stays.

17.3 Forecasting

Forecasting models could combine:

- historical UAC referrals;
- seasonal patterns;
- transfer rates;
- discharge rates;
- facility capacity;
- sponsor-processing times; and
- other operational predictors.

17.4 Scenario modeling

Decision-makers could simulate:

> What happens to HHS census if transfers increase by 20% while discharge throughput remains unchanged?

or:

> How quickly does system load decline if discharge throughput increases by 15%?

Such scenario analysis would make the dashboard more useful for contingency planning.

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18. Conclusion

This study demonstrates that UAC operational data can be transformed into a practical healthcare-capacity monitoring framework.

The analysis identifies three central characteristics of the observed system.

First, **HHS care represents the dominant component of active system workload**, substantially exceeding the CBP custody population on average.

Second, **system pressure is dynamic**. Although the overall observed average net pressure is negative, hundreds of reporting observations still experienced positive transfer-minus-discharge pressure. Consequently, aggregate averages alone cannot provide sufficient operational warning.

Third, **data quality is itself an operational issue**. The presence of 355 missing calendar dates and 86 transfer-versus-custody anomalies demonstrates the need to combine operational KPIs with reporting-quality indicators.

The recommended approach is therefore a continuous analytical framework built around:

System Load + Flow Pressure + Rolling Trends + Backlog Indicators + Data Quality.

Such a framework can help decision-makers move from retrospective reporting toward proactive operational awareness.

However, actual capacity strain should only be declared when workload indicators are compared against verified capacity information. Adding facility-level bed availability, staffing, utilization, placement time, and length-of-stay information would allow this framework to evolve from descriptive analytics into a full **predictive healthcare-capacity management system**.

Ultimately, the objective should not simply be to maximize throughput. In a child-centered care system, operational efficiency must remain aligned with safe, appropriate, trauma-informed, culturally and linguistically appropriate care and timely placement. ORR's policy direction toward flexible, community-based and family-based care reinforces the importance of viewing capacity as a continuum of care rather than simply as a count of available beds.

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19. Executive Summary of Findings

Area	Finding	Operational Meaning
Dataset	720 populated observations	Usable but incomplete time series
Reporting gaps	355 missing calendar dates	Do not interpret missing days as zero
Average system load	~6,233	Baseline workload indicator
Maximum system load	11,762	Historical peak
Average HHS care	~6,061	HHS is dominant workload component
Average CBP custody	~171	Smaller component of active load
Average daily transfers	~129	Incoming HHS flow
Average daily discharges	~173	Outflow generally offsets transfers
Positive-pressure observations	238	Periodic accumulation occurs
Negative-pressure observations	475	Relief periods are more common
Transfer anomalies	86	Require operational review
Discharge anomalies	0	Stronger internal consistency
Load volatility	~140 children	Meaningful daily variation

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20. Final Policy Recommendation

The strongest recommendation from this analysis is to establish a **UAC Care Load Early-Warning Dashboard** that combines historical census, current flow, rolling pressure, discharge throughput, and reporting quality.

The dashboard should not answer only:

> **"How many children are in care?"**

It should answer:

> **"How large is the current care population, is workload increasing or decreasing, how persistent is the pressure, how effectively is discharge activity offsetting incoming transfers, and how reliable is the reporting data?"**

That shift—from static counts to **dynamic care-system intelligence**—would provide substantially greater value for operational planning, staffing, shelter management, resource allocation, and humanitarian response.